

Wesley Rancher

Department of Geography
University of Oregon

[Terrestrial Ecosystems Ecology and Landscapes Lab](#)
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Summary

Environmental data scientist with expertise in geospatial science, remote sensing, and landscape ecology. Experienced in applying machine learning to large-scale spatial datasets to model ecosystem and carbon dynamics. Skilled at building reproducible workflows, integrating heterogeneous data, and communicating technical insights to both scientific and applied audiences. Quick learner, adaptable, and eager to apply analytical and programming skills in industry-focused geospatial projects.

Education

University of Oregon – M.S. in Geography, GPA: 3.9

Ohio Wesleyan University – B.A. in Environmental Studies and Geography, Minor: Philosophy, GPA: 3.5

Technical Skills

- **GIS & Remote Sensing:** ArcGIS Pro, QGIS, Google Earth Engine; LiDAR, multispectral
- **Programming:** Python (geopandas, rasterio, rioxtarray, scikit-learn), R (terra, tidymodels, randomForest), Bash, Docker, Git, AWS CLI
- **UAS:** FAA Part 107 Remote Pilot; sensor calibration and photogrammetry
- **Other:** Spatial statistics, data integration, workflow automation, reproducibility, mentoring/teaching
- **Languages:** Spanish (working proficiency)

Professional Experience

Graduate Research Assistant | University of Oregon

Sep 2023 – Present

Advisor: Dr. Melissa Lucash

- Built ETL pipelines combining Landsat imagery, topography, and climate projections to model aboveground biomass across boreal Alaska
- Developed Python and R workflows for raster/vector processing, random forest modeling, and automated analyses in Docker for reproducibility and scalability
- Managed and processed CMIP5/6 climate datasets for integration into predictive ecosystem models
- Collaborated with Bonanza Creek LTER to analyze forest change and wildfire impacts using remote sensing and simulation modeling
- Collaborated with USFS to use model carbon dynamics in Oregon Coast Range; converted large netCDF datasets to acceptable format for the LANDIS-II simulation model

Graduate Research Fellow | NASA DEVELOP

Jun 2023 – Aug 2023

Advisor: Dr. Anthony Vorster

- Mapped invasive vegetation in Utah using Landsat and LiDAR data to produce actionable maps and reports for stakeholders
- Integrated field data with vegetation indices and random forest models; automated analyses using ArcGIS and Google Earth Engine

Undergraduate Research Assistant | Ohio Wesleyan University

Dec 2022 – May 2023

Advisor: Dr. Nathan Rowley

- Modeled supraglacial lake depth in Western Greenland using radiative transfer models
- Developed reproducible Landsat workflows for raster sieving and feature detection

Teaching Experience

Graduate Teaching Assistant | University of Oregon

Sep 2023 – April 2025

- **Geography 485/585: Remote Sensing I:** Developed labs, taught GIS/remote sensing software (ArcGIS, QGIS, R), led hands-on demonstrations for undergraduates and graduates
- **Geography 199: Global Wildfire:** Guest lectures on “Changing Wildfire in Brazil” and “Bees and Wildfire”; supported curriculum development and supplemental instruction
- **Geography 181: Our Digital Earth:** Facilitated labs on digital mapping and spatial data; trained students in ArcGIS Online and coding workflows

Awards & Honors

- Rippey Research Grant (\$1,000) – University of Oregon
- NASA DEVELOP Scholarship (\$1,500) – SSAI
- Dean’s List – Ohio Wesleyan University
- Robert E. Shanklin Distinguished Scholar – Geography, OWU
- Phi Sigma Tau – Philosophy, OWU
- Our New Gold Digital Storytelling Winner – Spanish, OWU

Publications

- Rancher, W., Matsumoto, H., Lamping, J., & Lucash, M. (2025). *Estimating aboveground carbon using machine learning and process-based models. (In preparation)*
- Weiss, S., Rancher, W., Hayes, K., Buma, B., & Lucash, M. (2025). *Wildfire dynamics under climate change in interior Alaska. (In preparation)*
- Lucash, M., Lamping, J., Nowell, B., Scheller, R., Banerjee, T., Buettner, C., Fawcett, J., Hurteau, M., Parks, S., Rancher, W., Robbins, Z., St. Denis, L., Stasiewicz, A., Urza, A., & Weiss, S. (2025). *Roadmap for the future of extreme wildfire events. (Submitted)*
- Rowley, N., Rancher, W., & Karmosky, C. (2024). *Comparison of multiple methods for supraglacial melt-lake volume estimation in western Greenland during the 2021 summer melt season. Glaciers*

Presentations

- Rancher, W. (2025). *Estimating species-level aboveground carbon in interior Alaska using machine learning and process-based models. University of Oregon, Eugene, OR. (Master’s thesis presentation)*
- Rancher, W., Matsumoto, H., Lamping, J., & Lucash, M. (2025). *Estimating recent shifts in aboveground carbon and species composition in interior Alaska using Landsat imagery and random forests. Northwest Scientific Association, Eugene, OR. (Poster)*
- Rancher, W., Matsumoto, H., Lamping, J., & Lucash, M. (2024). *Assessing vegetation shifts in boreal Alaska by integrating Landsat imagery with spatial modeling. American Geophysical Union, Washington, DC. (Poster)*
- Rancher, W., VanArnam, M., Kowalski, A., Anarella, T., & Vorster, A. (2023). *Mapping Russian olive and tamarisk to inform invasive species management along the Paria River, Utah. NASA DEVELOP Day, Washington, DC. (Virtual talk)*
- Rancher, W., Rowley, N. (2023). *Estimating supraglacial melt lake volume changes in west-central Greenland using multiple remote sensing methods. Ohio Wesleyan Spring Symposium, Delaware, OH. (Poster)*